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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
Weekly Progress Report (WPR)

WPR Dates: 23/1/2026 - 30/1/2026 **WPR of Week:**2

Name: Santosh Kumar

Enrollment Number: 12196502722

Class & Section: CSE-8A

Department: Computer Science and Engineering

Faculty Supervisor's Name: Ms. Indu Sharma

MAJOR PROJECT TITLE: - "AURA : Autonomous User Realization & Assembly"

TARGETS SET FOR THE WEEK:

- Develop the first inventory model covering component type, value, quantity, and physical location.
- Study storage addressing strategies for drawers, boxes, and distributed shelves.
- Draft the communication flow between the ESP32 host and the distributed locator nodes.
- List the primary user actions for search, locate, and inventory checking.
- Prepare the initial component metadata structure for later circuit linkage.

ACHIEVEMENTS FOR THE WEEK:

- Designed the first version of the inventory record structure for value-aware and quantity-aware tracking.
- Mapped the basic physical storage addressing logic for racks, drawers, and grouped component rows.
- Prepared the host-to-node communication concept using addressed wireless triggering.
- Defined the main user actions for manual search, guided lookup, and part-location confirmation.
- Documented the initial metadata fields needed for component identity, value, quantity, and notes.

FUTURE WORK PLANS:

- Begin drafting the canonical JSON structure for circuit representation.
- Refine the digital-to-physical handoff flow from selected parts to locator activation.
- Start planning the host-side search interface and minimal control workflow.
- Review data needed for linking inventory with future circuit suggestions.

Week's Summary

Days/Time	5 Days a week Monday to Friday.
Monday	Defined the first inventory record structure covering part type, value, quantity, and location.
Tuesday	Worked on drawer and storage addressing logic for physically distributed component organization.
Wednesday	Outlined the host-to-node signaling model and locator-trigger workflow.
Thursday	Listed the main user actions for search, locate, confirm, and inventory review.
Friday	Compiled metadata requirements for linking physical stock with later circuit assistance.

STATUS OF RESEARCH PAPER: Literature review expanded with focus on inventory-aware hardware workflows and physical component retrieval systems. Estimated overall project completion by the end of this week: 19%.

Supervisor signature: **Date:**